IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Vavan Farizzaman

20 October 2026

Outline

Project question and executive summary

Data collection, preparation and analytical methods

Visualization and SQL findings

Folium maps and the Plotly Dash dashboard

Classification results, conclusions and technical appendix

Executive Summary

90 Falcon 9 launches support the main analysis. The course label marks 60 successful outcomes (66.7%).

Collection combines an API snapshot with documented enrichment. Scraping and the SQL/dashboard labs use separate datasets.

KSC LC-39A leads the dashboard sample: 10 successes from 13 launches (76.9%).

SVM, logistic regression and KNN each classify 15 of 18 test launches correctly. Training CV selects SVM.

Source: ML lab. Detailed references in speaker notes.

Introduction

Business context: first-stage recovery can support reuse and influence launch economics.

Question: which observed launch characteristics are associated with successful first-stage outcomes?

Approach: inspect payload, orbit, launch site and time, then compare four binary classifiers.

Scope: a historical learning exercise. Landing classification alone does not estimate launch cost or guarantee future recovery.

Source: Wrangling lab. Detailed references in speaker notes.

Section 1

Methodology

Methodology

1. Collect and reconcile launch data, resource IDs and source provenance.

2. Clean fields, label outcomes and encode 80 numerical features.

3. Explore patterns through Python charts, SQL queries and interactive maps.

4. Filter dashboard views by site and payload mass.

5. Tune four classifiers with 10-fold CV and evaluate a fixed 18-row test sample.

Data Collection

Source	Records	Use
API-derived Falcon 9 sample	90	Wrangling, EDA and classification
Wikipedia historical revision	121	Independent scraping exercise
Course SQL dataset	101	SQL aggregation and filters
Course map/dashboard datasets	56 each	Interactive analytics

Different scopes and labels produce different totals. These datasets are not pooled.

Source: API lab. Detailed references in speaker notes.

Data Collection: SpaceX API

Launch JSON supplies resource IDs, not all enrichment fields.

Archived API records and course reference data supply documented historical metadata.

Timeouts and caching keep requests explicit and repeatable.

Step	Output
1. Read launch snapshot	94 retained launch rows
2. Resolve resource IDs	Rockets, payloads, cores, pads
3. Keep Falcon 9	90 rows, excluding 4 Falcon 1
4. Export	dataset_part_1.csv

Source: API lab. Detailed references in speaker notes.

Data Collection: Web Scraping

Step	Method
1. Retrieve	Read the 9 June 2021 Wikipedia revision
2. Parse	BeautifulSoup extracts Falcon 9 launch tables
3. Normalize	Clean dates, payloads, sites and outcomes
4. Filter and export	121 numeric-flight rows in spacex_web_scraped.csv

The lab's numeric flight-number rule excludes Falcon Heavy rows labelled FH. Unknown payload masses remain missing.

Source: Scraping lab. Detailed references in speaker notes.

Data Wrangling

Stage	Implementation
Clean	Keep Falcon 9, normalize fields and impute missing payload mass with the mean
Label	Class 1: True ASDS, True RTLS or True Ocean. Other listed outcomes: class 0
Check	90 launches: 60 class 1 and 30 class 0. True ASDS contributes 41
Encode	One-hot encode orbit, site, landing pad and serial. Produce 80 float64 features

Outcome and Class are excluded from the feature matrix. Successful ocean outcomes remain included in the course target.

Source: Wrangling lab. Detailed references in speaker notes.

EDA with Data Visualization

Plot	Analytical purpose
Flight number vs. site and orbit	Inspect historical progression and category coverage
Payload vs. site and orbit	Compare operating ranges and overlapping outcomes
Success rate by orbit	Compare outcome proportions with sample counts
Yearly average success rate	Describe change over time without assuming causality

Color encodes Class. Counts accompany rates so small categories do not look as reliable as large samples.

Source: EDA lab. Detailed references in speaker notes.

EDA with SQL

SQLite stores 101 rows in a cleaned SPACEXTABLE, with dates and labels trimmed.

DISTINCT and LIKE identify sites. SUM and AVG aggregate payload mass.

MIN and MAX locate milestones and payload extremes. A subquery retains all maximum-payload boosters.

CASE separates mission outcomes. GROUP BY counts landing outcomes within an inclusive date window.

Source: SQL lab. Detailed references in speaker notes.

Interactive Map Methodology

Folium markers and circles identify launch sites. Popups preserve site names and counts.

Green/red markers show class 1/0. MarkerCluster groups overlapping launch coordinates.

Mouse coordinates, click popups and distance lines support proximity inspection.

OpenStreetMap features provide coastline, internal rail spur, major road and town-centre reference points.

Source: Folium lab. Detailed references in speaker notes.

Plotly Dash Methodology

A searchable launch-site dropdown filters both charts.

All Sites shows successful outcome counts by site. A selected site shows its success/failure split.

A 0-10,000 kg range slider filters the scatter plot, including both endpoints.

Scatter points use payload mass and class, with a consistent color for each booster category.

Source: Dash lab. Detailed references in speaker notes.

Predictive Analysis Methodology

Stage	Specification
Inputs	90 rows, 80 numerical features and binary Class labels
Holdout	72 train / 18 test. test_size=0.2, random_state=2
Models	Logistic regression, SVM, decision tree and KNN
Tuning	GridSearchCV, 10 folds, accuracy scoring
Scaling check	Repeat with StandardScaler inside each training-fold pipeline

The course sequence scales before splitting. The extra comparison addresses this scaling leakage while retaining the same test rows.

Source: ML lab. Detailed references in speaker notes.

Results Overview

Analysis	Finding
Python EDA, 90 launches	60 successful course labels. Orbit and year patterns differ
SQL, 101 launches	100 mission successes and 1 mission failure
Dashboard, 56 launches	KSC LC-39A leads in count and success rate
Classification, 18 test rows	Three methods tie at 83.3% accuracy

The following sections preserve each dataset's denominator and outcome definition.

Section 2

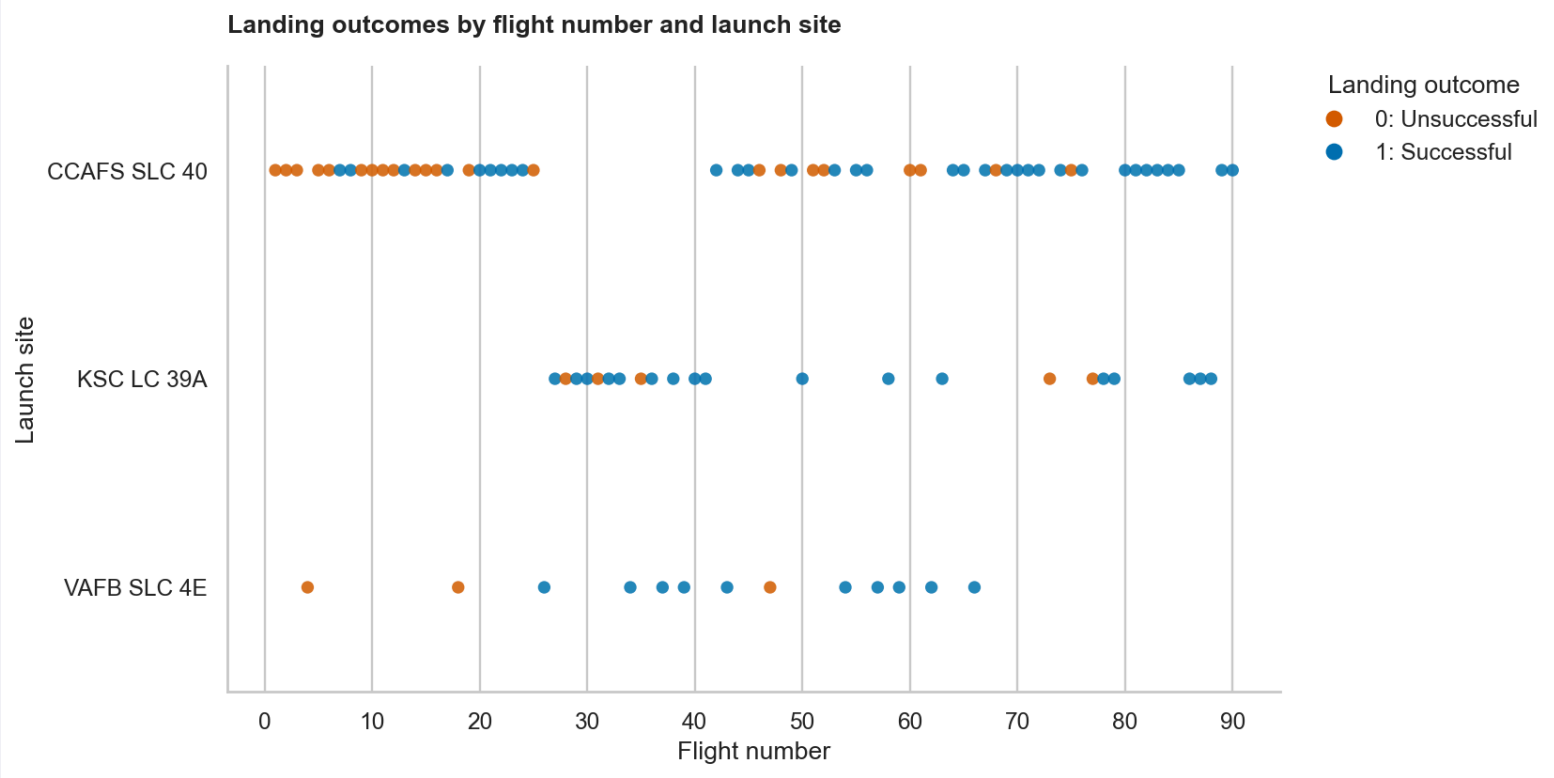
Insights drawn from EDA

Flight Number vs. Launch Site

CCAFS SLC 40 contributes 55 of 90 launches.

KSC: 17/22 successes.
VAFB: 10/13. CCAFS: 33/55.

Sites operate over different periods. A site effect is not isolated.



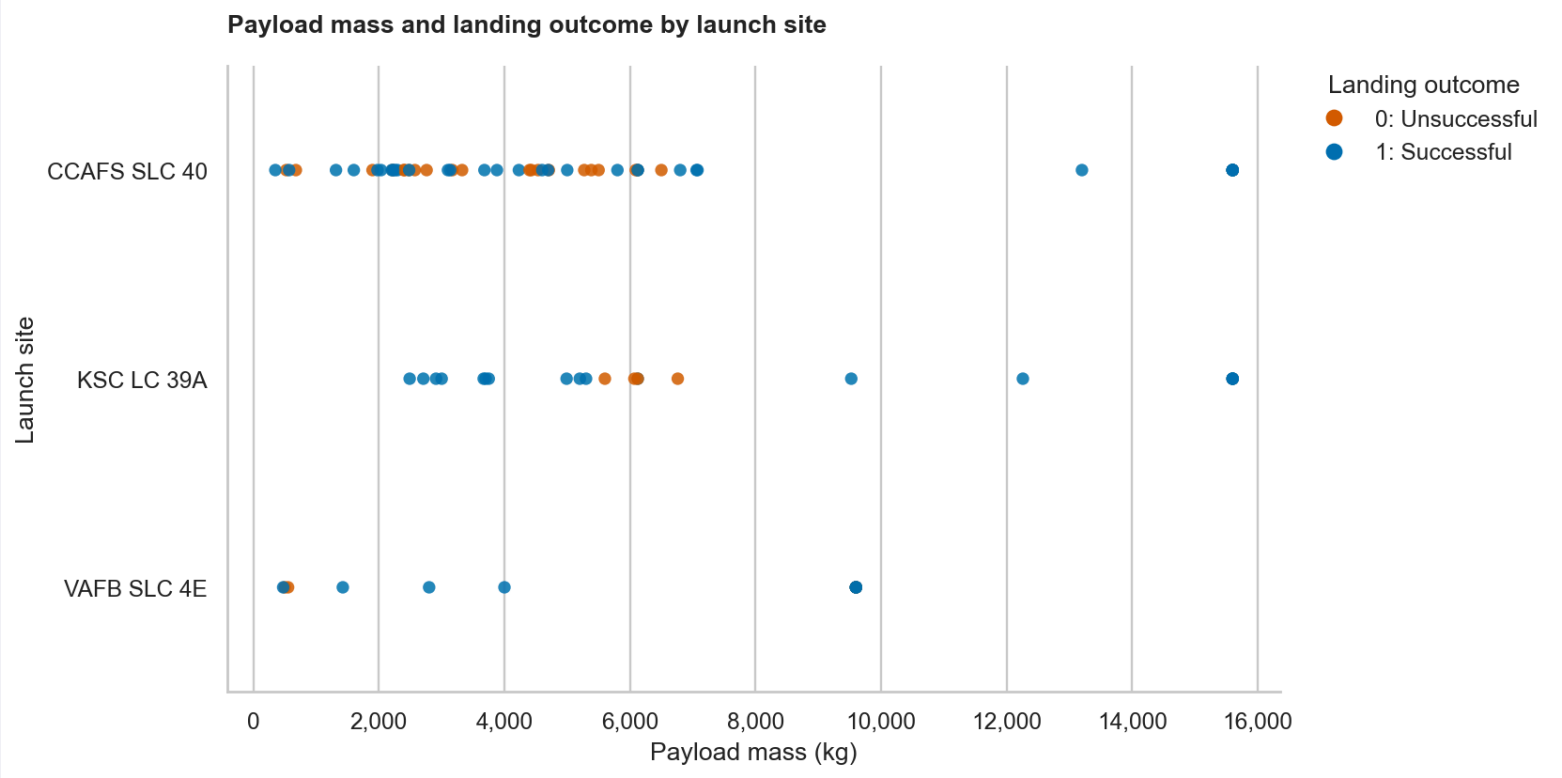
Source: EDA lab. Detailed references in speaker notes.

Payload vs. Launch Site

Payload ranges overlap across launch sites.

Both classes occur at similar masses. Mass alone does not separate outcomes.

Site comparisons should account for orbit and booster generation.



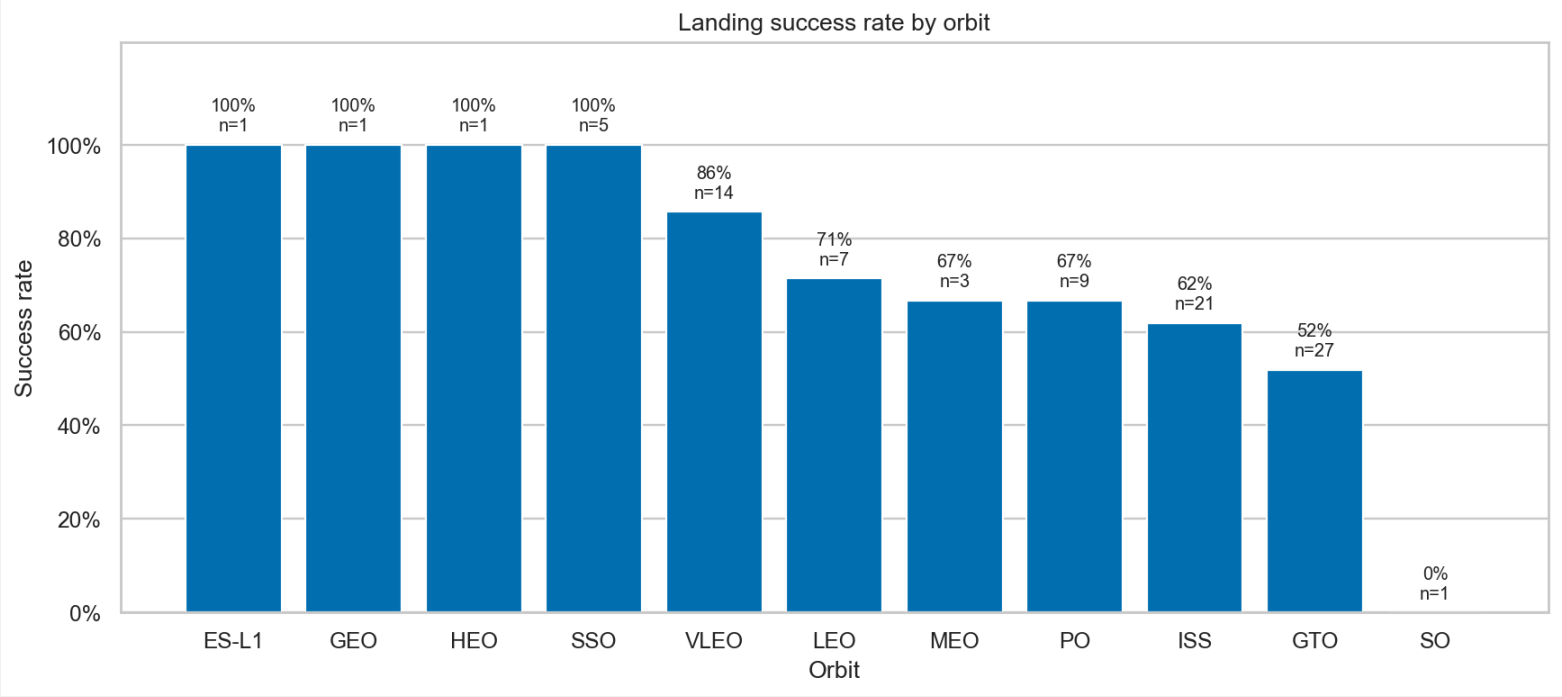
Source: EDA lab. Detailed references in speaker notes.

Success Rate by Orbit

VLEO: 12/14 successes (85.7%). GTO: 14/27 (51.9%).

SSO: 5/5. GEO, HEO and ES-L1 each have only one launch.

A 100% rate from one launch has very little supporting evidence.



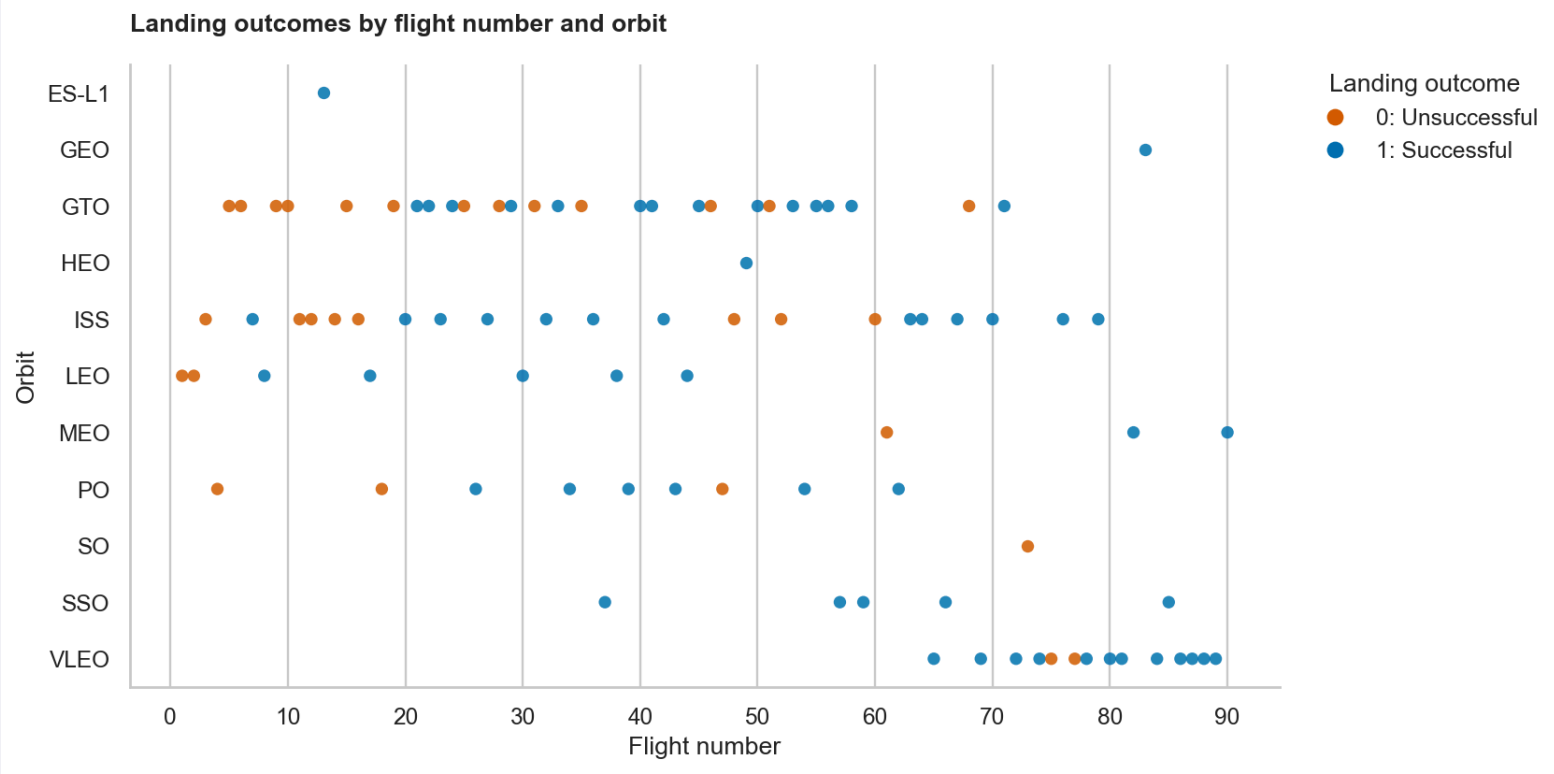
Source: EDA lab. Detailed references in speaker notes.

Flight Number vs. Orbit

Orbit categories occur at different stages of the launch history.

Later VLEO missions coincide with a more mature launch program.

Time and mission mix can confound an apparent orbit effect.



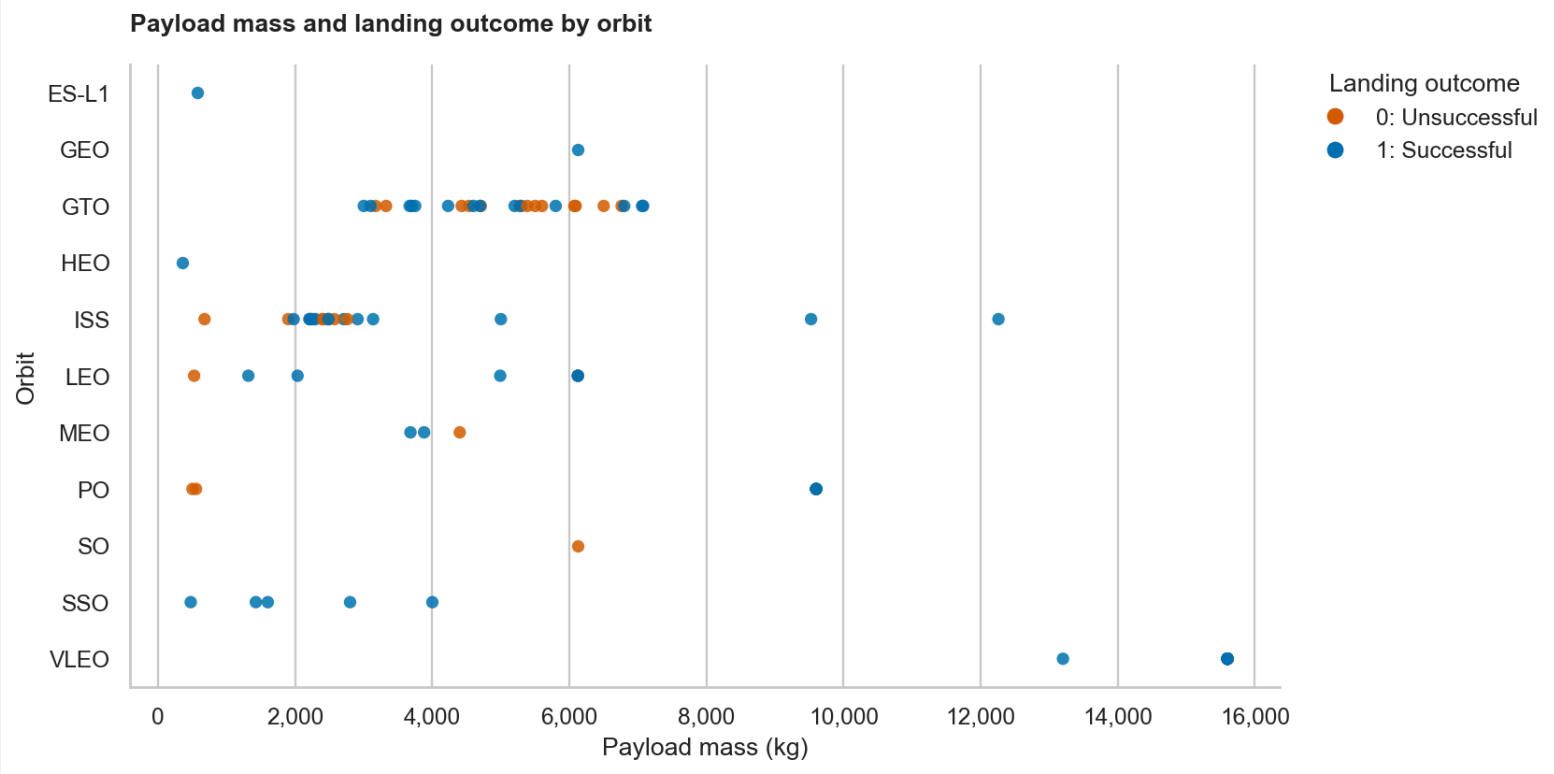
Source: EDA lab. Detailed references in speaker notes.

Payload vs. Orbit

Orbit constrains the payload range represented in the data.

GTO has 27 launches with mixed outcomes.

The scatter plot supports joint analysis of payload and orbit.



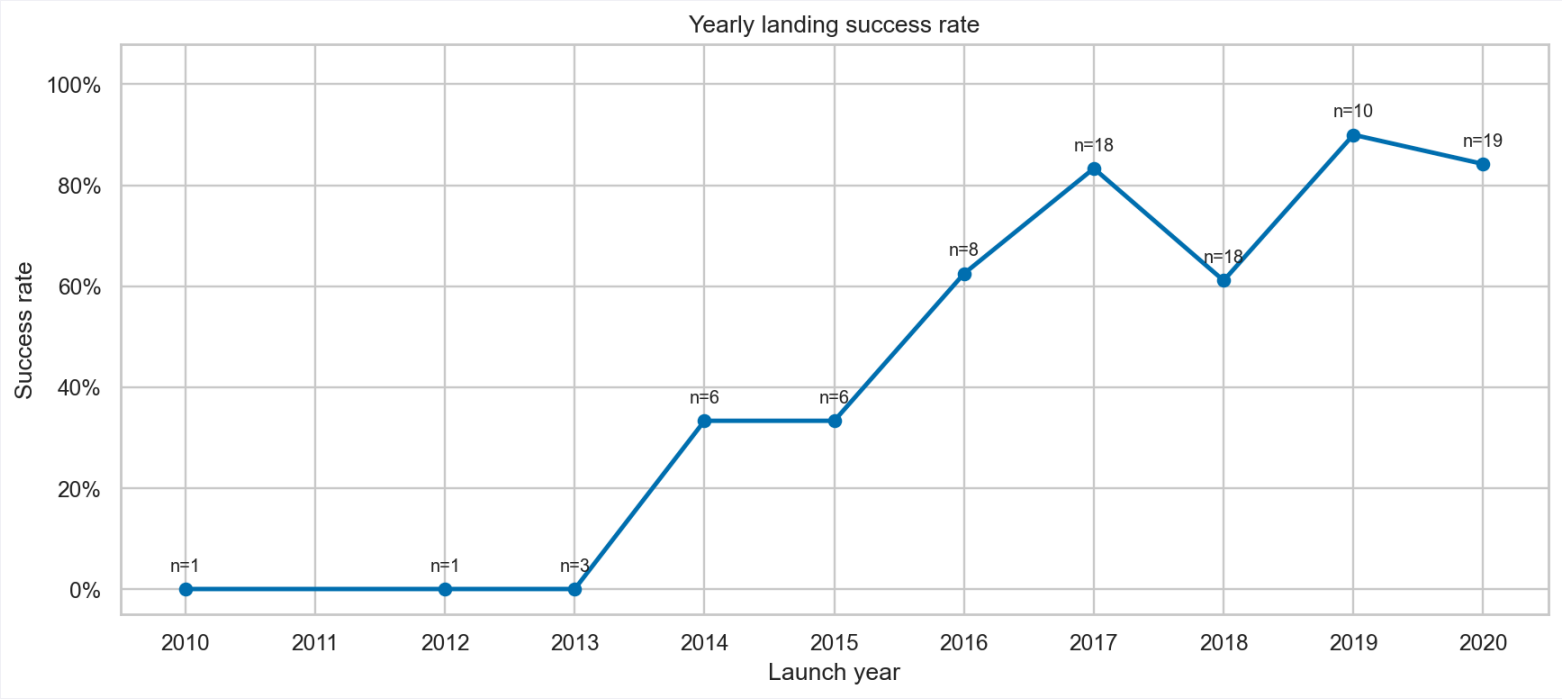
Source: EDA lab. Detailed references in speaker notes.

Yearly Success Trend

2013: 0/3. 2017: 15/18. 2020: 16/19 successful labels.

The rate generally improves but drops to 11/18 in 2018.

No 2011 record appears in this sample. Missing years are not zero rates.



Source: EDA lab. Detailed references in speaker notes.

Unique Launch Sites

```
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE ORDER BY Launch_Site;
```

Launch site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Four labels appear in the SQL source. This spelling differs from the 90-row API-derived sample.

Source: SQL lab. Detailed references in speaker notes.

First Five Records Matching CCA

```
SELECT Date, Booster_Version, Launch_Site
FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%'
ORDER BY Date, "Time (UTC)" LIMIT 5;
```

Date	Booster	Launch site
2010-06-04	F9 v1.0 B0003	CCAFS LC-40
2010-12-08	F9 v1.0 B0004	CCAFS LC-40
2012-05-22	F9 v1.0 B0005	CCAFS LC-40
2012-10-08	F9 v1.0 B0006	CCAFS LC-40
2013-03-01	F9 v1.0 B0007	CCAFS LC-40

An explicit chronological order makes LIMIT 5 reproducible.

Source: SQL lab. Detailed references in speaker notes.

Total Payload for NASA (CRS)

```
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass_KG
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Customer filter	Total payload mass
NASA (CRS), exact label	45,596 kg

This total covers the NASA (CRS) category only. It does not combine other NASA customer labels.

Source: SQL lab. Detailed references in speaker notes.

Average Payload for F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass_KG
FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

Booster filter	Average payload mass
F9 v1.1, exact label	2,928.4 kg

Serial-number variants are separate strings and do not match this exact filter.

Source: SQL lab. Detailed references in speaker notes.

First Successful Ground-Pad Landing

```
SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

Landing outcome	Earliest date in dataset
Success (ground pad)	22 December 2015

The result follows the SQL dataset's date field.

Source: SQL lab. Detailed references in speaker notes.

Drone-Ship Successes at 4,000-6,000 kg

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Matching booster versions
F9 FT B1021.2
F9 FT B1031.2
F9 FT B1022
F9 FT B1026

Strict bounds exclude payloads equal to 4,000 or 6,000 kg.

Source: SQL lab. Detailed references in speaker notes.

Mission Outcomes in the SQL Dataset

```
SELECT CASE WHEN Mission_Outcome LIKE 'Success%' THEN 'Success'
WHEN Mission_Outcome LIKE 'Failure%' THEN 'Failure' END AS Result,
COUNT(*) FROM SPACEXTABLE GROUP BY Result;
```

Mission outcome	Count	Share
Success	100	99.0%
Failure	1	1.0%

Mission success concerns the mission objective. Booster landing outcomes are a different field.

Source: SQL lab. Detailed references in speaker notes.

Boosters Carrying the Maximum Payload

```
SELECT DISTINCT Booster_Version FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

Maximum payload mass: 15,600 kg. Twelve distinct booster-version records match.

Booster version	Booster version
F9 B5 B1048.4	F9 B5 B1051.4
F9 B5 B1048.5	F9 B5 B1051.6
F9 B5 B1049.4	F9 B5 B1056.4
F9 B5 B1049.5	F9 B5 B1058.3
F9 B5 B1049.7	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1060.3

Source: SQL lab. Detailed references in speaker notes.

Drone-Ship Landing Failures in 2015

```
SELECT Date, Booster_Version, Launch_Site FROM SPACEXTABLE
WHERE SUBSTR(Date, 1, 4) = '2015'
AND Landing_Outcome = 'Failure (drone ship)' ORDER BY Date;
```

Month	Booster	Launch site
January	F9 v1.1 B1012	CCAFS LC-40
April	F9 v1.1 B1015	CCAFS LC-40

Both records have Landing_Outcome = Failure (drone ship).

Source: SQL lab. Detailed references in speaker notes.

Landing Outcomes: June 2010 to March 2017

```
SELECT Landing_Outcome, COUNT(*) FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC;
```

Landing outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

31 launches match the inclusive date window. Counts are in descending order.

Source: SQL lab. Detailed references in speaker notes.

Section 4

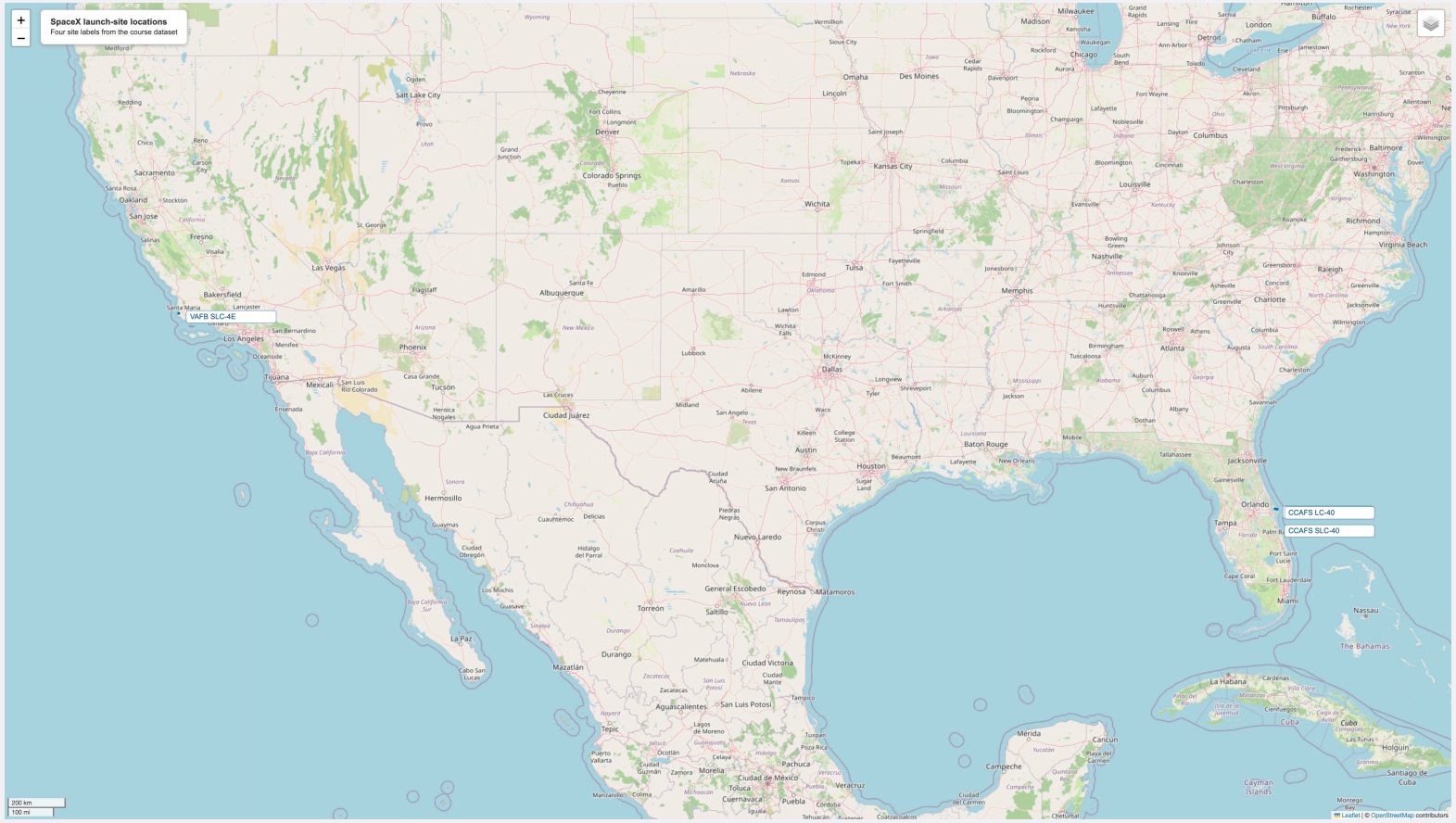
Launch Sites Proximities Analysis

Launch Sites on the US Coasts

The course data identifies four labels at coastal launch facilities.

VAFB is in California. The other three labels are in Florida.

Nearby Florida sites overlap at this map scale.



Source: Folium lab. Detailed references in speaker notes.

Launch Outcomes Behind the Folium Markers

Launch site	Launches	Successes	Success rate
CCAFS LC-40	26	7	26.9%
CCAFS SLC-40	7	3	42.9%
KSC LC-39A	13	10	76.9%
VAFB SLC-4E	10	4	40.0%

The map encodes each launch as green (class 1) or red (class 0). Clusters group repeated site coordinates. This table makes the underlying rates explicit.

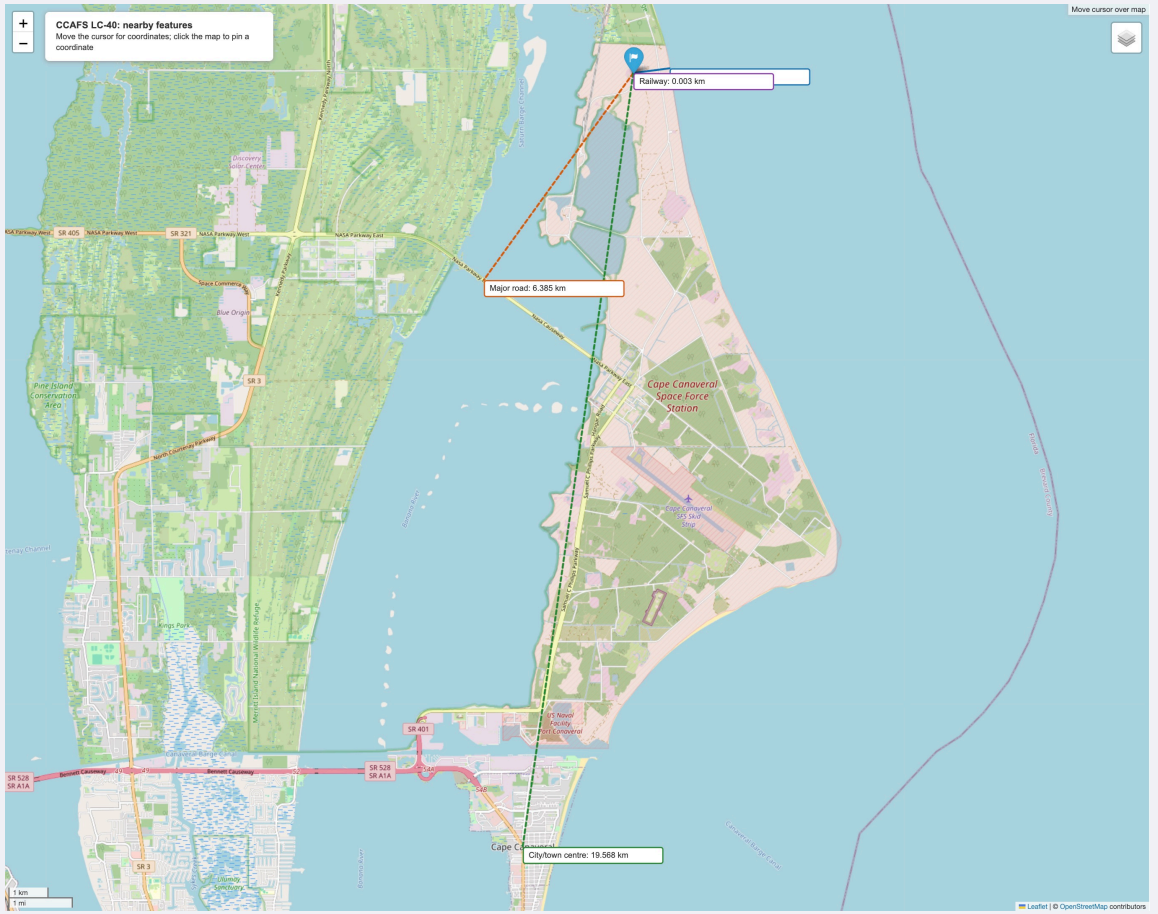
Source: Folium lab. Detailed references in speaker notes.

CCAFS LC-40 Proximity Analysis

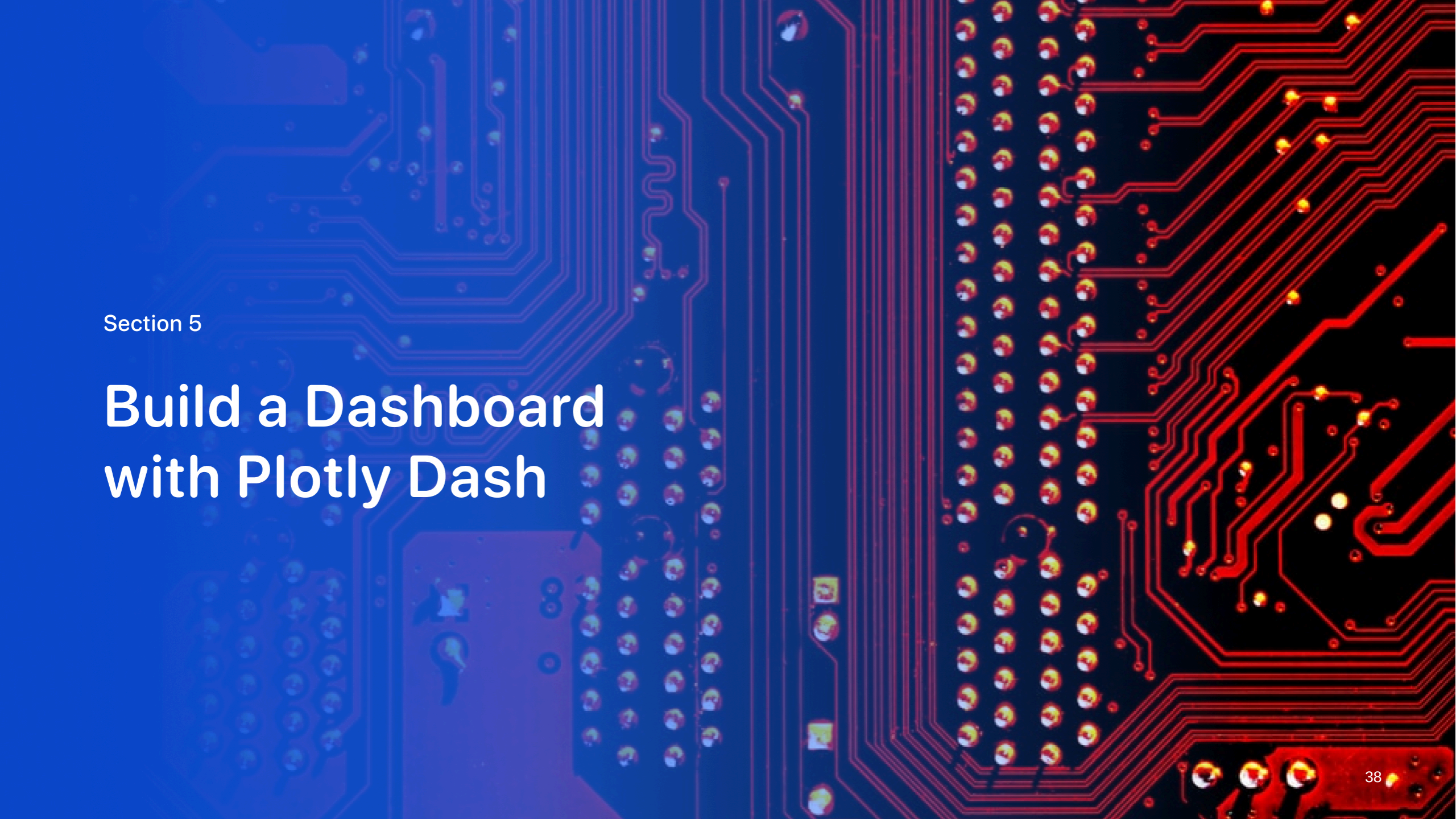
Coastline: 0.922 km. Major road: 6.385 km.

Cape Canaveral centre: 19.568 km. Internal rail spur: about 3 m.

Distances use mapped reference points. Rail is a local spur, not the external network.



Source: Folium lab. Detailed references in speaker notes.



Section 5

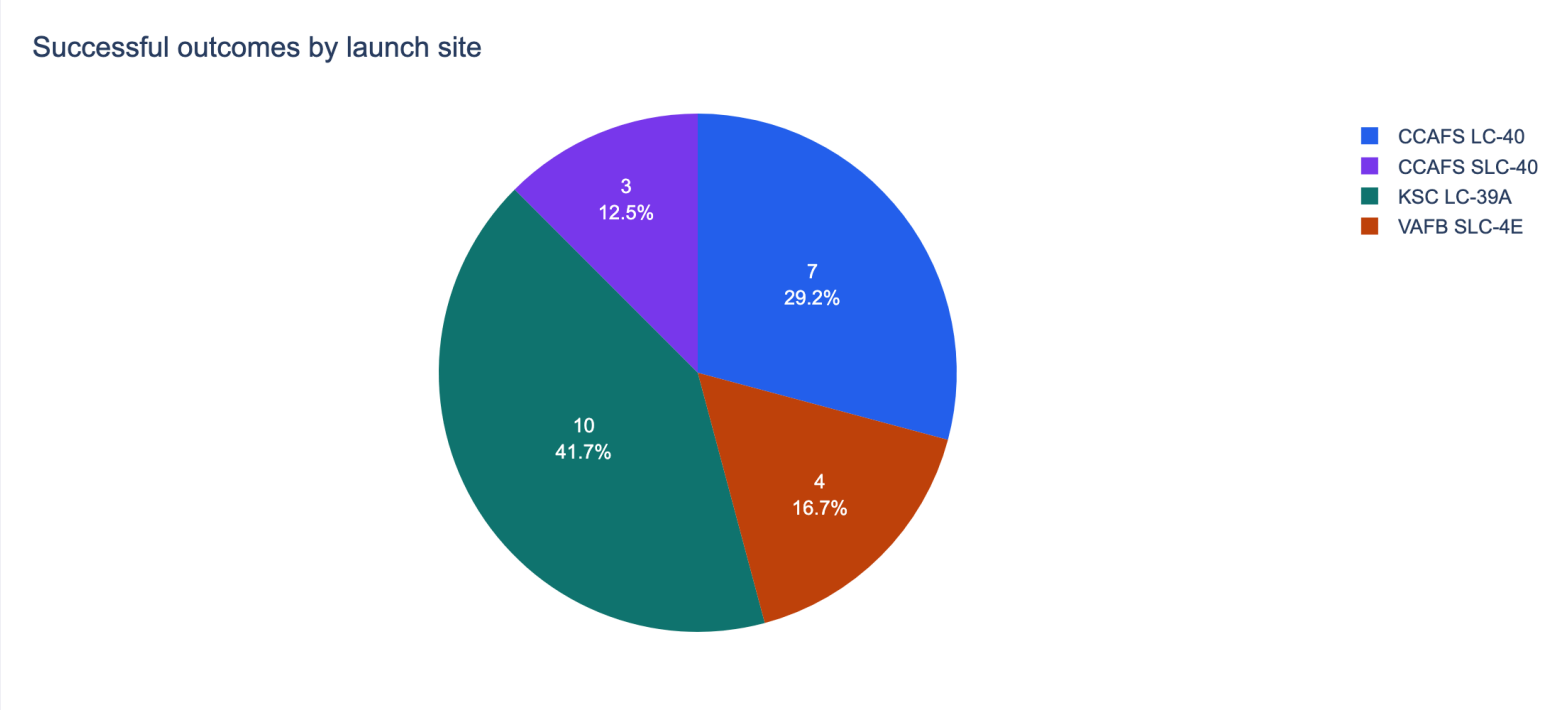
Build a Dashboard with Plotly Dash

Successful Outcomes Across All Sites

The 56-row dashboard sample contains 24 successful outcomes.

KSC LC-39A contributes 10, or 41.7% of all successes.

A pie share is different from a site's success rate.



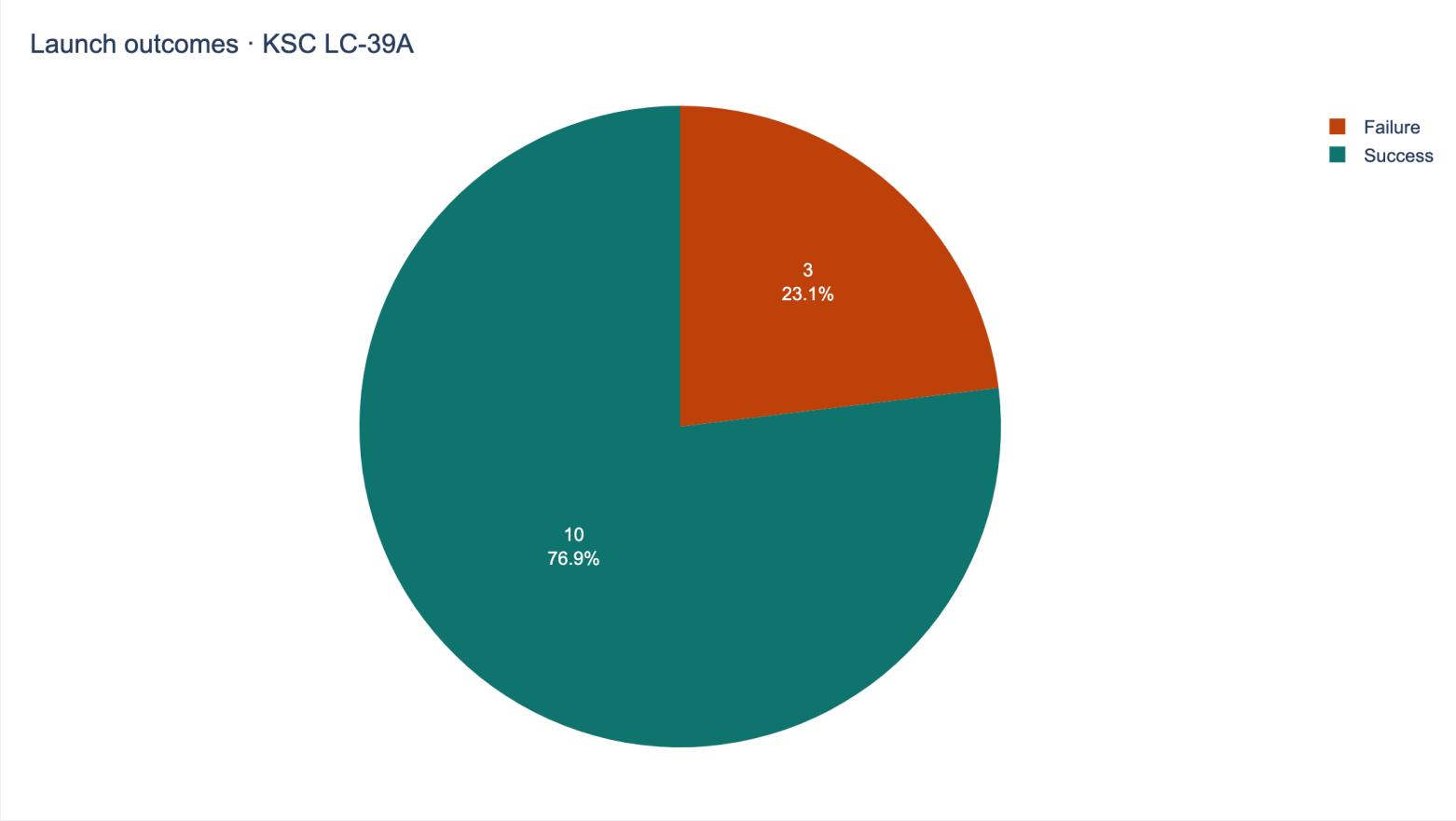
Source: Dash lab. Detailed references in speaker notes.

KSC LC-39A Has the Highest Site Rate

10 successful and 3 unsuccessful outcomes.

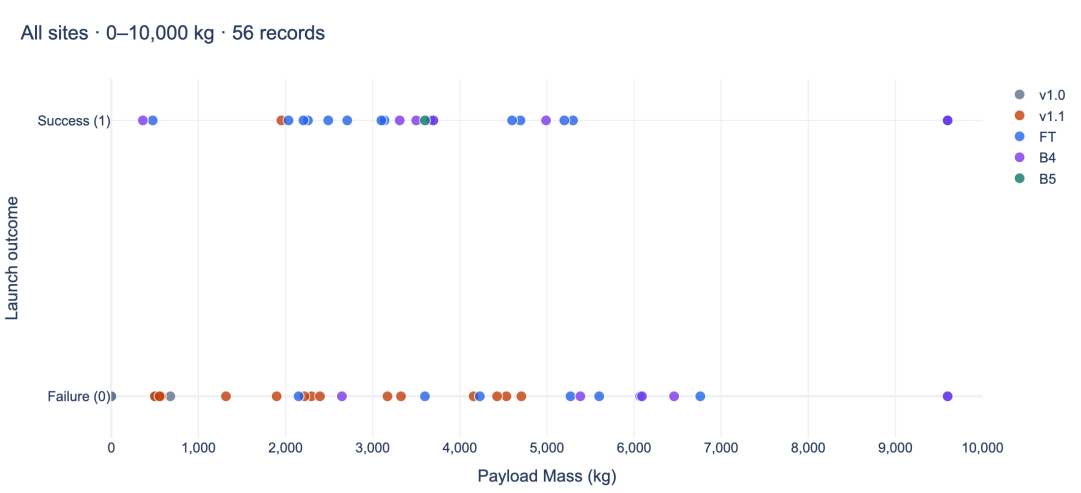
Success rate: 10/13 = 76.9%, the highest in this sample.

The selected-site pie changes the denominator to that site's launches.

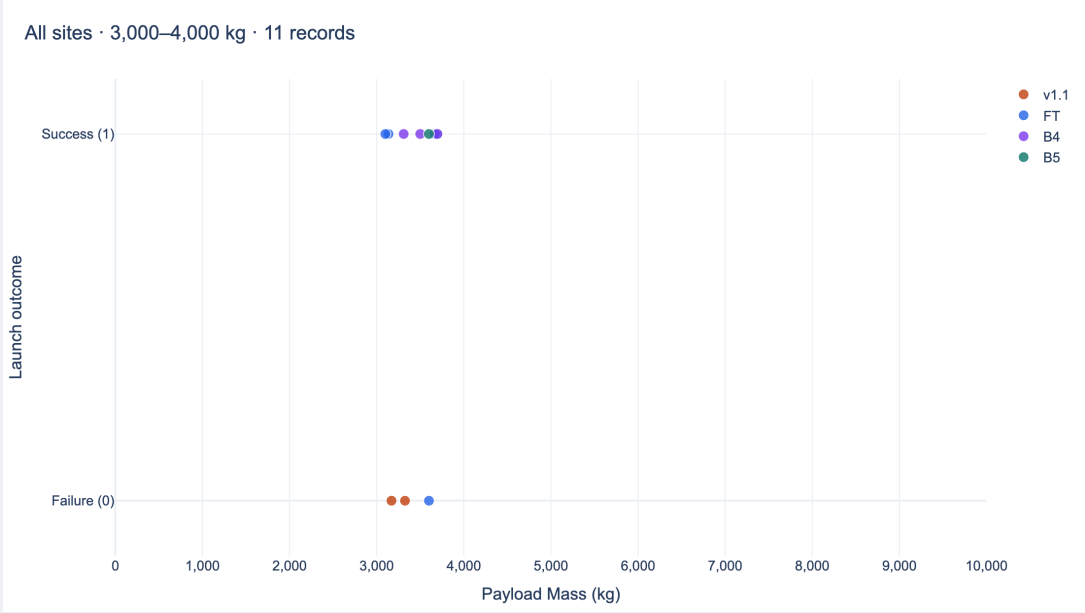


Source: Dash lab. Detailed references in speaker notes.

Payload Filters Reveal Different Outcome Mixes



Full payload range



Filtered payload range

Using fixed 1,000 kg bins: 3,000 to under 4,000 kg has 8/11 successes (72.7%). The 6,000 to under 7,000 kg bin has 0/4.

B5 has 1/1 success. FT has 16/24, providing more evidence than a single B5 observation.

Source: Dash lab. Detailed references in speaker notes.

Section 6

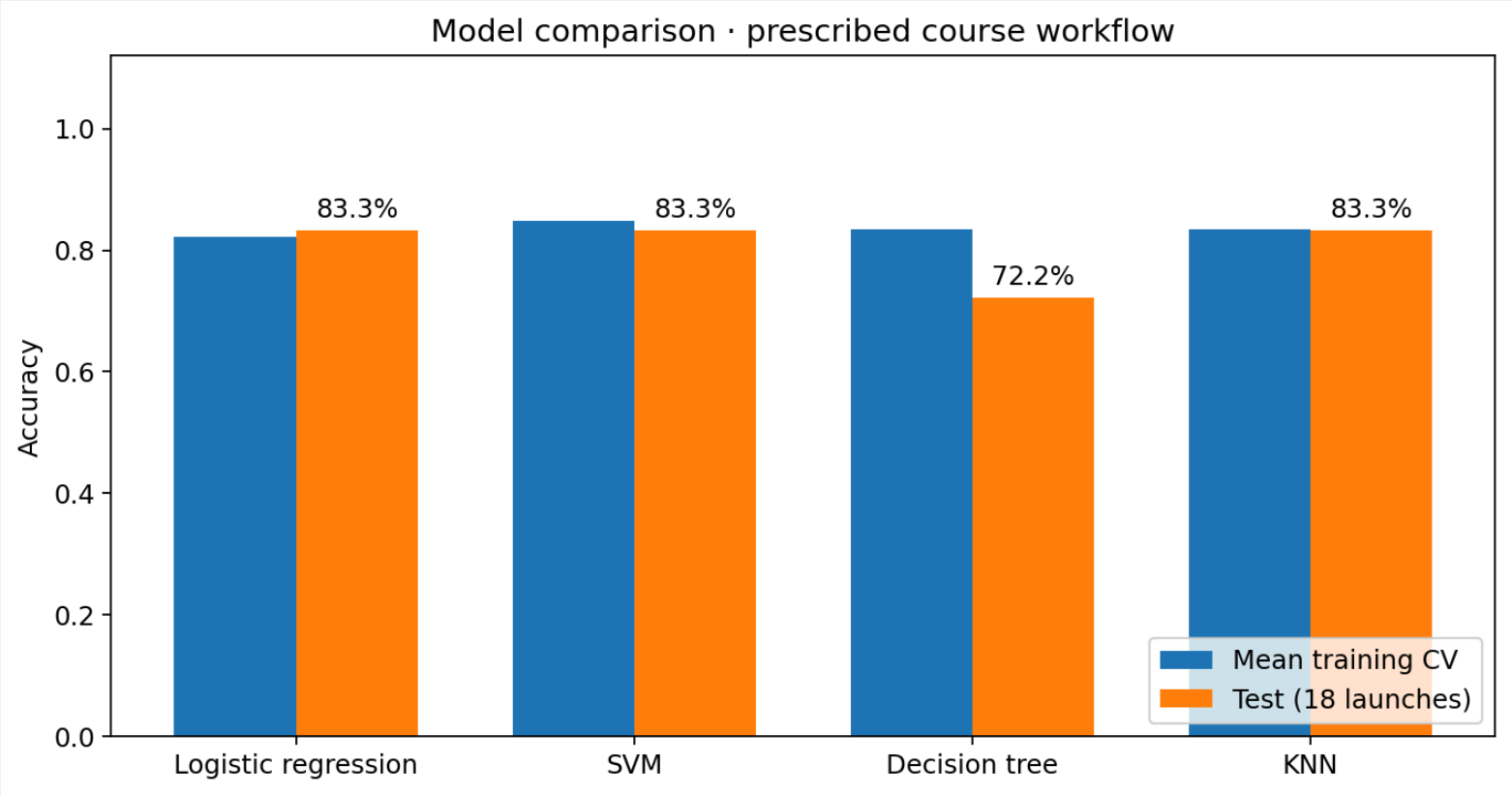
Predictive Analysis (Classification)

Classification Accuracy

Logistic regression, SVM and KNN tie: 15/18 correct (83.3%).

Decision tree: 13/18 (72.2%). Majority baseline: 12/18 (66.7%).

Fold-fitted scaling selects SVM by CV: 83.6% CV, 83.3% test.



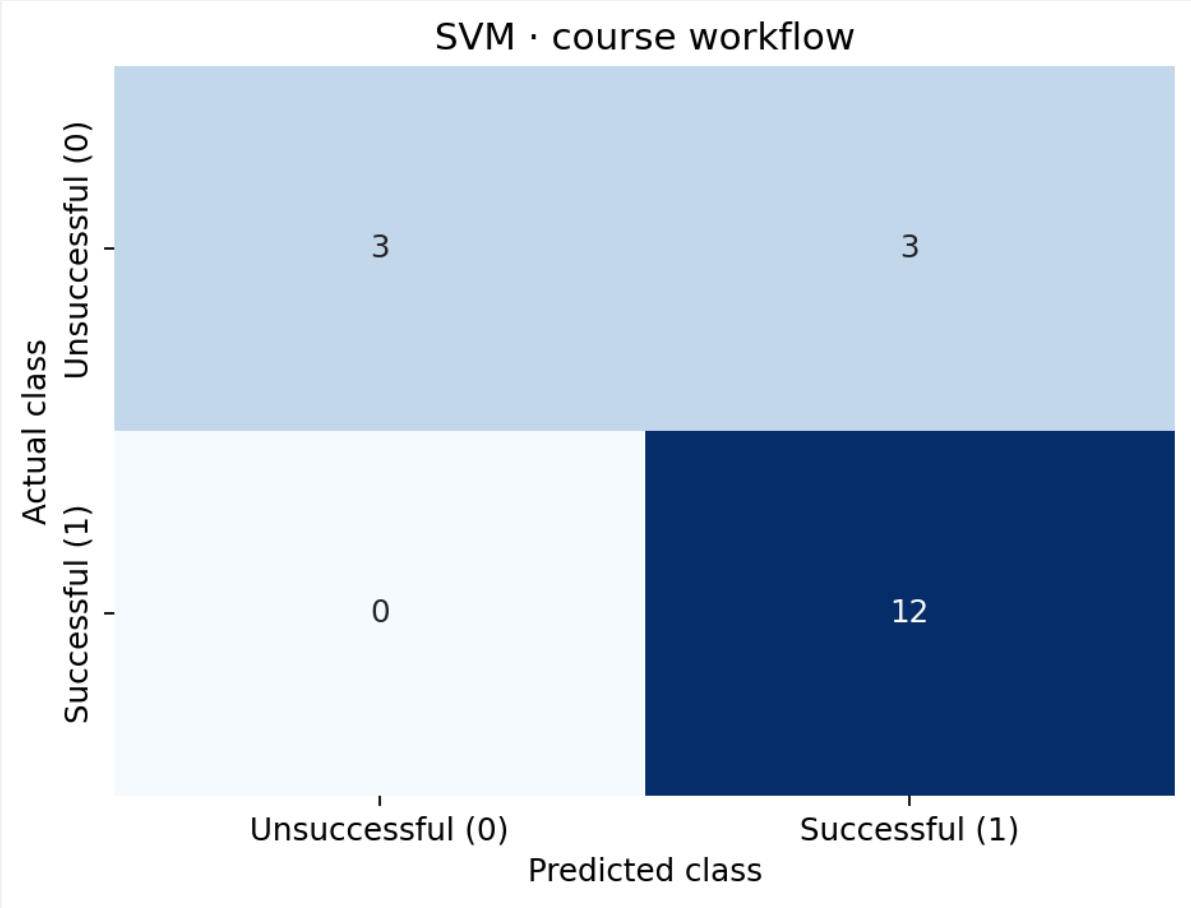
Source: ML lab. Detailed references in speaker notes.

SVM Confusion Matrix

True negatives: 3. False positives: 3. False negatives: 0. True positives: 12.

Success recall: 100%.
Success precision: 80%.
Specificity: 50%.

All three errors predict success for an unsuccessful launch.



Source: ML lab. Detailed references in speaker notes.

Conclusions

Historical outcome rates differ by orbit, year and site. Their effects overlap and are not causal estimates.

The dashboard's strongest site is KSC LC-39A at 10/13 successful outcomes.

Three classifiers achieve 83.3% test accuracy. Training cross-validation favors SVM.

Next steps: use a future-time holdout, rebuild preprocessing on training data, and audit feature availability before launch.

Source: ML lab. Detailed references in speaker notes.

Appendix: Dataset and Label Register

Analysis	Rows	Positive outcome definition
API / EDA / ML	90	True ASDS, True RTLS or True Ocean
SQL	101	Mission success and landing outcome are separate fields
Folium / Dash	56 each	Supplied binary class in each course file
Wikipedia scraping	121	Source outcome text, not merged into ML labels

The main 90-row sample has 41 successful drone-ship outcomes. Including 14 ground and 5 ocean successes produces 60 positive labels.

Additional Insights and Evaluation Limits

Observation	Implication
15 correct predictions out of 18	Approximate 95% Wilson interval: 60.8%-94.2%
3 of 6 class-0 launches predicted as class 1	Accuracy conceals a 50% false-positive rate for this class
Five True Ocean outcomes count as positive	Excluding ocean outcomes gives 55/90 = 61.1%, not 66.7%
Scaling inside CV changes model tuning	KNN chooses k=5 instead of k=6, despite the same test accuracy

These checks sharpen the interpretation. They do not justify claiming deployment readiness.

Source: ML lab. Detailed references in speaker notes.

Code and Reproducibility

GitHub project repository

https://github.com/VavanZaa/IBM-Capstone

Artifact	Local location
Completed notebooks and Python files	capstone/
Dataset snapshots and provenance	capstone/data/
Charts, maps and model predictions	dataviz_outputs/, location_outputs/, ml_outputs/
Dashboard entry point	spacex_dash_app.py
Run instructions and dependencies	RUN_INSTRUCTIONS.txt and requirements.txt

Reproduce the dashboard with Python and rerun notebooks from top to bottom. This report uses the saved historical inputs.

Sources and Technical References

IBM Skills Network course datasets: API_call_spacex_api.json, dataset_part_1.csv, Spacex.csv, spacex_launch_geo.csv and spacex_launch_dash.csv.

Wikipedia historical launch tables: revision 1027686922, dated 9 June 2021.

Archived SpaceX API metadata: resource URLs and field provenance in data/spacex_lab_metadata.json.

OpenStreetMap contributors: extracted features dated 20 October 2026. ODbL attribution applies to the map data.

scikit-learn: model selection and preprocessing guidance. Plotly Dash and Folium: interactive visualization libraries.

Thank you!